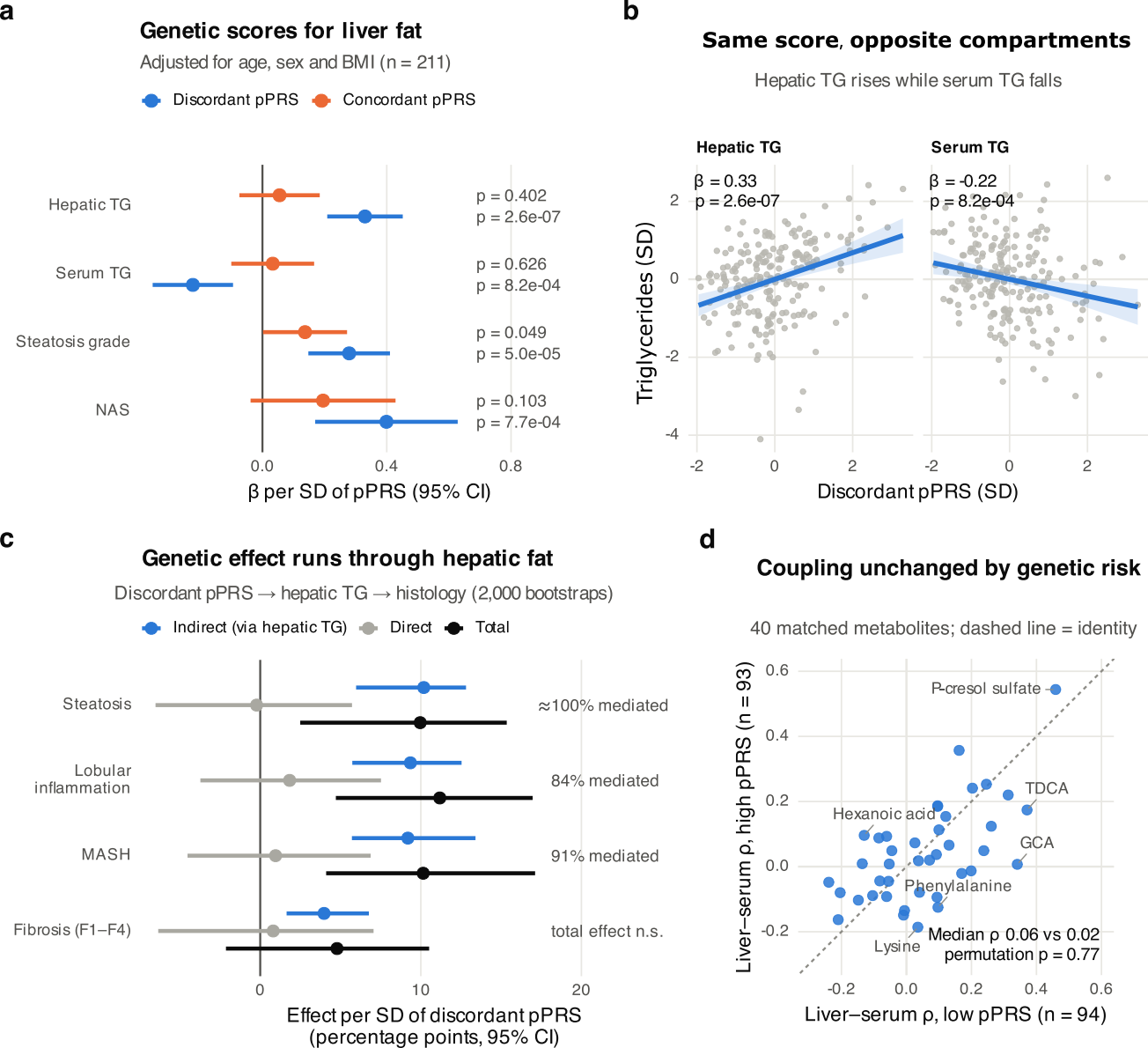

a
Genetic scores for liver fat
Adjusted for age, sex and BMI (n = 211)
Discordant pPRS
Concordant pPRS
Hepatic TG
Serum TG
Steatosis grade
NAS
p = 0.402
p = 2.6e-07
p = 0.626
p = 8.2e-04
p = 0.049
p = 5.0e-05
p = 0.103
p = 7.7e-04
0.0
0.4
0.8
β per SD of pPRS (95% CI)
b
Same score, opposite compartments
Hepatic TG rises while serum TG falls
Hepatic TG
Serum TG
β = 0.33
p = 2.6e-07
β = -0.22
p = 8.2e-04
Triglycerides (SD)
Discordant pPRS (SD)
c
Genetic effect runs through hepatic fat
Discordant pPRS → hepatic TG → histology (2,000 bootstraps)
Indirect (via hepatic TG)
Direct
Total
Steatosis
Lobular inflammation
MASH
Fibrosis (F1–F4)
≈100% mediated
84% mediated
91% mediated
total effect n.s.
Effect per SD of discordant pPRS
(percentage points, 95% CI)
d
Coupling unchanged by genetic risk
40 matched metabolites; dashed line = identity
P-cresol sulfate
TDCA
Hexanoic acid
GCA
Phenylalanine
Lysine
Median ρ 0.06 vs 0.02
permutation p = 0.77
Liver–serum ρ, high pPRS (n = 93)
Liver–serum ρ, low pPRS (n = 94)